Reads over Atom Spaces: A Mathematical View of Context, Retrieval, and Parameter Routing in Language Models

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Research Highlights

- One read operator over atom spaces covers attention, retrieval, and MoE routing
- A Lipschitz insertion condition and an exact tail certificate are shown to be far too loose in practice
- Anti-atom cancellation is exact but collapses under key perturbations of order 0.05
- Held-out ROC calibration lifts CounterFact-500 paraphrase activation from 38% to 53%
- Matched GRACE run: 99–100% rewrite efficacy but 0–6% paraphrase activation on GPT-2

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